

CEBE Interdisciplinary Fellowship Programme 2026

Curriculum Vitae

1. Personal and Academic Information

Name: Christos T. Georgakis

Title: Professor

Affiliation: Aarhus University

Research group: Structural Dynamics and Geotechnical Engineering; CEBE RF7 – Extending the Life of the Built Environment

PhD year: 2001

ORCID: 0000-0001-6001-3856

2. Research Achievements and Relevant Impact

Research achievements:

Christos T. Georgakis is Professor of Structural Dynamics and Monitoring at Aarhus University and Lead of CEBE Research Field 7, *Extending the Life of the Built Environment*. His research spans structural health monitoring, bridge engineering, structural dynamics, wind engineering, vibration control and environmental loading, with particular emphasis on methods that enable safer and longer use of existing infrastructure. His work has combined experimental mechanics, full-scale monitoring, wind-tunnel testing, operational modal analysis, virtual sensing and, more recently, machine-learning-based prediction and nowcasting of extreme environmental events. He is co-author of *Cable Supported Bridges: Concept and Design* and has led or participated in numerous interdisciplinary and industry-funded research programmes.

Impact:

His research has been applied to major international infrastructure including the London Millennium Bridge, Queensferry Crossing, Port Mann Bridge and Gordie Howe International Bridge, and he led the independent investigation of the Chirajara Bridge collapse. His work has also generated patented vibration- and cable-control technologies, including solutions licensed to industry. In 2026 he received the **Robert H. Scanlan Medal** from ASCE's Engineering Mechanics Institute for extraordinary contributions to bridge, aerodynamics, and wind engineering.

Relevance to the proposed project:

Within CEBE RF7, Georgakis contributes senior expertise in structural health monitoring, predictive monitoring of extreme events, infrastructure dynamics and industry translation. His role is particularly relevant to projects that use sensing, modelling and prediction to support condition assessment, risk-informed intervention and safe life extension of existing structures.

3. Supervision and Research Training

<i>Category</i>	<i>Current</i>	<i>Former</i>
PhD students	3	19
Postdoctoral researchers	0	8
Master's students	1	80

4. Selected Publications

A. Five Most Important Publications — Last 15 Years

1. **Gimsing, N.J. & Georgakis, C.T.** *Cable Supported Bridges: Concept and Design*, 3rd ed. Wiley, 2012. DOI: 10.1002/9781119978237.
2. **Tarpø, M.G., Nabuco, B., Georgakis, C.T. & Brincker, R.** “Expansion of experimental mode shape from operational modal analysis and virtual sensing for fatigue analysis using the modal expansion method.” *International Journal of Fatigue*, 130, 105280, 2020. DOI: 10.1016/j.ijfatigue.2019.105280
3. **Demartino, C., Koss, H.H., Georgakis, C.T. & Ricciardelli, F.** “Effects of ice accretion on the aerodynamics of bridge cables.” *Journal of Wind Engineering and Industrial Aerodynamics*, 138, 98–119, 2015. DOI: 10.1016/j.jweia.2014.12.010.
4. **Chen, J. & Georgakis, C.T.** “Tuned rolling-ball dampers for vibration control in wind turbines.” *Journal of Sound and Vibration*, 332(21), 5271–5282, 2013. DOI: 10.1016/j.jsv.2013.05.019.
5. **Maienza, C., Avossa, A.M., Ricciardelli, F., Coiro, D., Troise, G. & Georgakis, C.T.** “A life cycle cost model for floating offshore wind farms.” *Applied Energy*, 266, 114716, 2020. DOI: 10.1016/j.apenergy.2020.114716.

B. Five Recent Publications Relevant to the Proposed Project — Last 5 Years

6. **Georgakis, C.T., Lagaros, N.D., Demartino, C. & Modé, P.D.** “Predictive Monitoring of Extreme Events for Enhanced Bridge Safety Using Hyperlocal ML-Powered Wind Nowcasting.” *e-Journal of Nondestructive Testing*, 31(2), 2026. DOI: 10.58286/32477. **Project relevance:** Predictive monitoring and short-term extreme-event decision support for infrastructure.
7. **Wei, J., Narazaki, Y., Quaranta, G., Yang, Q., Georgakis, C.T. & Demartino, C.** “Wind profile nowcasting and forecasting using machine learning.” *Journal of Wind Engineering and Industrial Aerodynamics*, 266, 106162, 2025. DOI: 10.1016/j.jweia.2025.106162. **Project relevance:** ML-based environmental sensing and prediction for structural and infrastructure applications.
8. **Modé, P., Demartino, C., Georgakis, C.T. & Lagaros, N.D.** “Short-term extreme wind speed forecasting using dual-output LSTM-based regression and classification model.” *Journal of Wind Engineering and Industrial Aerodynamics*, 259, 106035, 2025. DOI: 10.1016/j.jweia.2025.106035. **Project relevance:** Data-driven prediction of rare environmental hazards affecting infrastructure safety and operation.
9. **Kristoffersen, J.C., Kabel, T., Georgakis, C.T., Bellos, V. & Karmpadakis, I.** “Spatio-temporal measurement of laboratory wave fields using LiDAR.” *Coastal Engineering*, 209, 105047, 2026. DOI: 10.1016/j.coastaleng.2026.105047. **Project relevance:** High-resolution, non-contact sensing and environmental load characterisation.
10. **Matejčka, L., Delpech, P., Georgakis, C.T. & Flamand, O.** “An experimental investigation of ice and snow shedding from bridge cables with and without surface modifications.” *Cold Regions Science and Technology*, 220, 104148, 2024. DOI: 10.1016/j.coldregions.2024.104148. **Project relevance:** Monitoring, understanding and mitigation of climate-related hazards affecting bridge operation and service life.

Relevant industry/public-sector partnerships: *International bridge engineering, monitoring and forensic assignments include the Gordie Howe International Bridge, Queensferry Crossing, Port Mann Bridge, Wittpenn Bridge and the independent investigation of the Chirajara Bridge collapse.*

Selected awards/recognition: *2026 Robert H. Scanlan Medal*, American Society of Civil Engineers, Engineering Mechanics Institute, *2009 Reinholdt W. Jorck and Family Foundation Career Achievement Award*.